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├── package.json                # Project dependencies
├── tsconfig.json              # TypeScript configuration
├── wxt.config.ts              # WXT build system config
├── jest.config.ts             # Jest test configuration
├── playwright.config.ts       # Playwright E2E
configuration
├── .eslintrc.json            # ESLint configuration
├── .prettierrc               # Prettier formatting
├── src/types/                # TypeScript type
definitions
├──   ├── torrent.ts          # Torrent data types
├──   ├── config.ts           # Configuration types
├──   └── api.ts              # API response types
├──   ├── src/shared/         # Shared utilities
├──   │   ├── constants.ts    # Regex patterns, constants
├──   │   ├── logger.ts       # Structured logging
├──   │   ├── errors.ts       # Custom error classes
├──   │   ├── events.ts       # Typed event bus
├──   │   ├── crypto.ts       # AES-256-GCM encryption
├──   │   ├── storage.ts      # chrome.storage wrapper
├──   │   └── utils.ts        # Debounce, throttle,
utilities
├──   ├── src/parser/         # Torrent parsing engine
├──   │   ├── magnet.ts       # Magnet link parser
├──   │   ├── bencode.ts      # Bencode encoder/decoder
├──   │   └── torrent-file.ts # .torrent file parser
├──   ├── src/scanner/        # DOM scanning engine
├──   │   ├── base.ts         # Abstract scanner
├──   │   ├── link-scanner.ts # Link element scanner
├──   │   ├── text-scanner.ts # Text node scanner
├──   │   ├── site-db.ts      # Site-specific selectors
├──   │   └── orchestrator.ts # Scanner orchestrator
├──   ├── src/api/            # API integration layer
├──   │   ├── client.ts       # Boba API client
├──   │   ├── auth.ts         # Authentication handlers
├──   │   ├── qbittorrent.ts  # qBitTorrent API adapter
├──   │   ├── health.ts       # Health checker
├──   │   └── queue.ts         # Offline queue system
├──   ├── src/background/     # Service Worker
├──   │   └── index.ts         # Background script entry
├──   ├── src/content/        # Content Script
├──   │   ├── index.ts        # Content script entry
├──   │   ├── scanner.ts      # DOM scanning logic
├──   │   ├── highlight.ts     # Visual overlay
├──   │   └── styles.css       # Content script styles

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└─ src/popup/	# Popup UI
└─ index.html	# Popup HTML
└─ popup.ts	# Popup logic
└─ styles.css	# Popup styles
└─ src/options/	# Options Page
└─ index.html	# Options HTML
└─ options.ts	# Options logic
└─ styles.css	# Options styles
└─ src/assets/	# Icons and assets
└─ icon.svg	# SVG icon source
└─ locales/en/	# Internationalization
└─ messages.json	# English strings
└─ tests/	# COMPREHENSIVE TEST SUITES
└─ unit/	# Unit tests (Jest)
└─ magnet.test.ts	# Magnet parser tests (50+)
└─ bencode.test.ts	# Bencode tests (50+)
└─ torrent-file.test.ts	# Torrent file tests (40+)
└─ scanner.test.ts	# Scanner tests (45+)
└─ api-client.test.ts	# API client tests (50+)
└─ queue.test.ts	# Queue tests (45+)
└─ crypto.test.ts	# Encryption tests (40+)
└─ chrome-mock.ts	# Browser API mocks
└─ setup.ts	# Test setup
└─ fixtures/	# Test fixtures
└─ magnets.json	# Magnet URI test fixtures
└─ e2e/	# E2E tests (Playwright)
└─ popup.spec.ts	# Popup E2E tests (20+)
└─ content.spec.ts	# Content script E2E (25+)
└─ options.spec.ts	# Options E2E tests (35+)
└─ .github/workflows/	# CI/CD pipelines
└─ ci.yml	# Lint, test, build, E2E
└─ release.yml	# Release to all stores
└─ sql/	# SQL schemas
└─ schemas.sql	# Full database schema
└─ migrations/001_initial.sql	# Initial migration
└─ diagrams/	# 36 DIAGRAMS IN 5 FORMATS
└─ 01-system-architecture.*	# System architecture (5 files)
└─ 02-dataflow-torrent-detection.*	# Data flow (3 files)
└─ 03-dataflow-tab-groups.*	# Tab group batch flow (3 files)
└─ 04-sequence-single-torrent.*	# Single torrent sequence (3 files)
└─ 05-sequence-tab-group-batch.*	# Batch sequence (3 files)

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|   |─ 06-component-diagram.*          # Component diagram (3
files)
|   |─ 07-er-diagram.*                # ER diagram (4 files)
|   |─ 08-extension-lifecycle.*        # Lifecycle (3 files)
|   |─ 09-offline-queue-state-machine.* # State machine (2 files)
|   |─ 10-security-architecture.*      # Security (3 files)
|   |─ 11-build-pipeline.*             # CI/CD pipeline (3 files)
|   |─ 12-compatibility-matrix.*       # Browser matrix (2 files)
|
|─ docs/                               # DOCUMENTATION (25,725+
words)
|   |─ technical-specification.md       # Technical spec (5,433
words)
|   |─ api-reference.md                # API reference (5,999
words)
|   |─ user-guide.md                  # User guide (5,390 words)
|   |─ developer-guide.md              # Developer guide (4,800
words)
|   |─ installation-guide.md           # Installation guide (4,103
words)
|
|─ final/                              # MULTI-FORMAT
DOCUMENTATION
|   |─ technical-specification.pdf      # PDF format (521 KB)
|   |─ technical-specification.docx     # DOCX format (34 KB)
|   |─ technical-specification.html     # HTML format (61 KB)
|   |─ api-reference.pdf               # PDF format (482 KB)
|   |─ api-reference.docx              # DOCX format (42 KB)
|   |─ api-reference.html              # HTML format (77 KB)
|   |─ user-guide.pdf                  # PDF format (449 KB)
|   |─ user-guide.docx                 # DOCX format (30 KB)
|   |─ user-guide.html                 # HTML format (56 KB)
|   |─ developer-guide.pdf             # PDF format (422 KB)
|   |─ developer-guide.docx            # DOCX format (35 KB)
|   |─ developer-guide.html            # HTML format (60 KB)
|   |─ installation-guide.pdf          # PDF format (427 KB)
|   |─ installation-guide.docx         # DOCX format (29 KB)
|   |─ installation-guide.html         # HTML format (49 KB)
|
|─ research/                           # DEEP RESEARCH (27,560
lines)
|   |─ boba_extension_dimensions.md     # Dimension decomposition
|   |─ boba_extension_cross_verification.md # Cross-verification
results
|   |─ boba_extension_insight.md        # Strategic insights (10
insights)
|   |─ boba_extension_dim01.md          # Dimension 01: Boba
Architecture
|   |─ boba_extension_dim02.md          # Dimension 02: qBitTorrent
API
|   |─ boba_extension_dim03.md          # Dimension 03: Extension
Architecture

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└─ boba_extension_dim04.md	# Dimension 04: Cross-
Browser Compat	
└─ boba_extension_dim05.md	# Dimension 05: Tab Groups
API	
└─ boba_extension_dim06.md	# Dimension 06: Magnet
Parsing	
└─ boba_extension_dim07.md	# Dimension 07: Torrent
Parsing	
└─ boba_extension_dim08.md	# Dimension 08: DOM
Scanning	
└─ boba_extension_dim09.md	# Dimension 09: API
Integration	
└─ boba_extension_dim10.md	# Dimension 10: UI/UX
Design	
└─ boba_extension_dim11.md	# Dimension 11: Security
Model	
└─ boba_extension_dim12.md	# Dimension 12: Testing &
Build	

File Count Summary

Category	Files	Formats
Source Code	60	TypeScript, HTML, CSS, JSON, YAML, SQL
Diagrams	36	Mermaid, SVG, draw.io, HTML, PlantUML
Documentation (source)	5	Markdown
Documentation (PDF)	5	PDF
Documentation (DOCX)	5	DOCX
Documentation (HTML)	5	HTML
Research	15	Markdown
Plans	2	Markdown
TOTAL	133	8 formats

Key Statistics

- **Total research:** 27,560 lines across 12 dimensions

- **Source code:** 60 production-ready TypeScript/HTML/CSS files
 - **Test coverage:** 320+ test cases (unit + E2E)
 - **Documentation:** 25,725+ words across 5 documents
 - **Diagrams:** 36 files across 12 diagram types and 5 formats
 - **Total package size:** ~4.8 MB
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Quick Start for Development Team

1. **Read the Implementation Plan:** `implementation-plan.md` — 8 phases with fine-grained tasks
 2. **Review Technical Spec:** `docs/technical-specification.md` — architecture, requirements, data models
 3. **Explore Source Code:** `src/` — complete, compilable TypeScript
 4. **Run Tests:** `cd src && npm test` (after `npm install`)
 5. **Build Extension:** `cd src && npm run build`
 6. **Review Diagrams:** `diagrams/` — all architecture and flow diagrams
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Verification Checklist

- ☒ Deep research (12 dimensions with cross-verification)
- ☒ Implementation plan (all phases, sub-phases, tasks)
- ☒ Complete source code (TypeScript, strict mode, production-ready)
- ☒ SQL schemas (tables, indexes, migrations)
- ☒ Unit tests (320+ test cases, 80%+ coverage target)
- ☒ E2E tests (Playwright, 80+ test cases)
- ☒ CI/CD pipelines (GitHub Actions)
- ☒ Technical specification (25 FRs, 15 NFRs)
- ☒ API reference (all endpoints with curl/JS/TS examples)
- ☒ User guide (step-by-step with troubleshooting, 30 FAQs)

- ☒ Developer guide (setup, build, test, contribute)
 - ☒ Installation guide (all 5 browsers)
 - ☒ System architecture diagrams
 - ☒ Data flow diagrams
 - ☒ Sequence diagrams
 - ☒ Component diagrams
 - ☒ ER diagrams
 - ☒ State machine diagrams
 - ☒ Security architecture diagrams
 - ☒ Cross-browser compatibility matrix
 - ☒ All diagrams in Mermaid, SVG, PNG reference, draw.io, UML
 - ☒ Documentation in Markdown
 - ☒ Documentation in PDF (enterprise styled)
 - ☒ Documentation in DOCX (enterprise styled)
 - ☒ Documentation in HTML (clean styled)
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This package was generated on 2026-06-06 and represents a complete, production-ready deliverable for the BobaLink browser extension project.